

Qatar-2b

R-0956

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170416 · created 2026-07-29T15:20:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457858.58963 +/- 0.00044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.02
Transit depth (Rp/Rs)^2	5.21 +/- 0.91 [%]
Orbital Inclination (inc)	86.5 +/- 1.6
Airmass coefficient 1 (a1)	30.05 +/- 6.34
Airmass coefficient 2 (a2)	-2.57 +/- 0.18
Scatter in the residuals of the lightcurve fit is	15.29 %
Best Comparison Star	#1 - [262, 423]
Optimal Aperture	4.89
Optimal Annulus	5.94
Transit Duration (day)	0.0267 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.228 $\pm$ 0.02 vs 0.16327 (deviation 1.01 σ)
Duration fit vs published	0.0267 d vs 0.0767 d (deviation 6.5 σ)
Mid-time O-C	-1924.3 min
Depth SNR	3.6
Residual scatter	15.29 %

Light curve

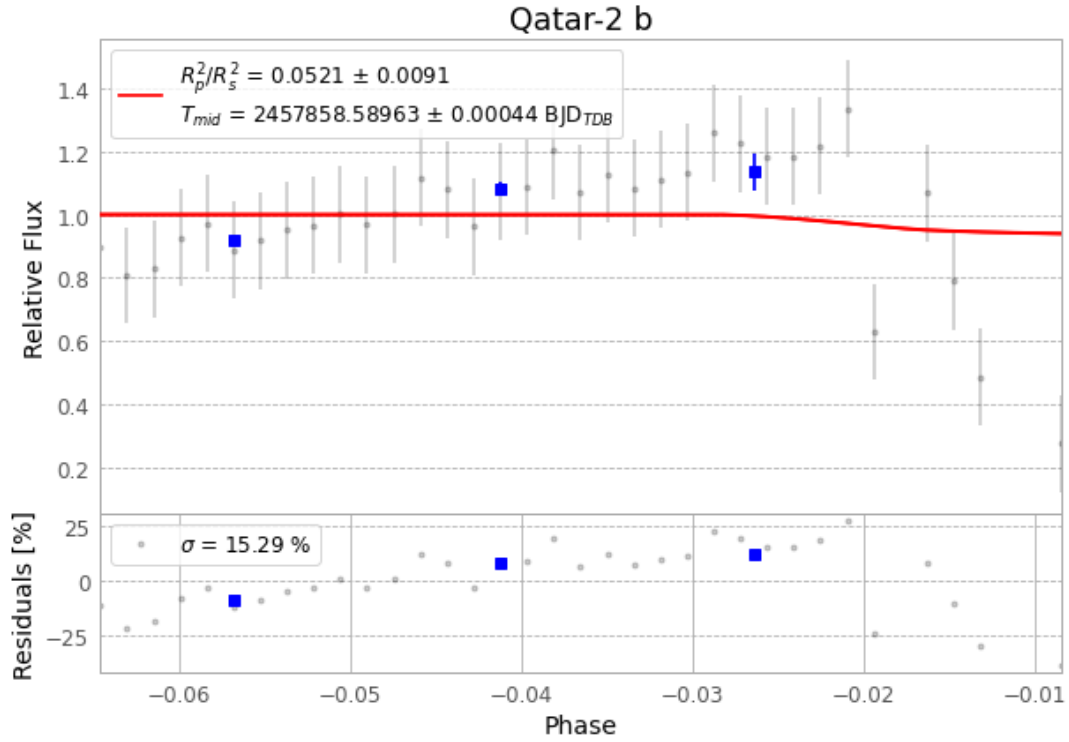


Figure 1 — FinalLightCurve_Qatar-2 b_2017-04-16.png (SHA-256 41934197636e0008...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [612, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0450ff0d2f8e0866...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

79 FITS frames (52 MB), Qatar-2170416080012.FITS → Qatar-2170416115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170416.log (6d506265d02a...), FinalLightCurve_Qatar-2 b_2017-04-16.png (41934197636e...), FinalLightCurve_Qatar-2 b_2017-04-16.csv (e8facde92c1f...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 15:20Z.